NORMALIZATION EXAMPLES

ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

Additional material from Connolly

Update Anomalies Recap

- Tables that contain redundant information may potentially suffer from update anomalies
- Types of update anomalies include:
  - Insertion – how do you insert details of a new distribution center that has no employees?
  - Deletion – when we delete the last employee in a distribution center, we lose the information about the distribution center
  - Modification – changes to a distribution center must be made for all records containing that distribution center

Normal Forms

- First Normal Form
  - All rows must have the same number of columns
  - Single valued attributes only
- Third Normal Form
  - Every column is fully functionally dependent on the primary key

Example

- The DB table below shows the parking space and location assigned to each employee
  - What are the update anomalies (insert, delete, and update)

ID	staffNo	staffName	garage	location	capacity	floor	spaceNo
1	S1156	Jane Jones	Yellow	Block E	120	3	123
2	S2311	Karen Gilmore	Yellow	Block E	120	3	145
3	S1167	Richard Blight	Yellow	Block E	120	3	156
4	S2345	Guy Richie	Green	Block D	45	2	26
5	S3434	Stephen Williams	Green	Block D	45	2	34

Update to 1NF

- The staffName attribute is not single valued

ID	staffNo	staffName	garage	location	capacity	floor	spaceNo
1	S1156	Jane Jones	Yellow	Block E	120	3	123
2	S2311	Karen Gilmore	Yellow	Block E	120	3	145
3	S1167	Richard Blight	Yellow	Block E	120	3	156
4	S2345	Guy Richie	Green	Block D	45	2	26
5	S3434	Stephen Williams	Green	Block D	45	2	34

1NF

- The staffName attribute is now single valued by separating the name field into two fields

ID	staffNo	staffFirstName	staffLastNam	garage	location	capacity	floor	spaceNo
1	S1156	Jane	Jones	Yellow	Block E	120	3	123
2	S2311	Karen	Gilmore	Yellow	Block E	120	3	145
3	S1167	Richard	Blight	Yellow	Block E	120	3	156
4	S2345	Guy	Richie	Green	Block D	45	2	26
5	S3434	Stephen	Williams	Green	Block D	45	2	34
#####						0	0	0

3NF Errors

- Notice that staffNo, staffFirstName, staffLastName, and spaceNo depend on ID
- Notice that location, capacity, and floor depend on garage

ID	staffNo	staffFirstName	staffLastNam	garage	location	capacity	floor	spaceNo
1	S1156	Jane	Jones	Yellow	Block E	120	3	123
2	S2311	Karen	Gilmore	Yellow	Block E	120	3	145
3	S1167	Richard	Blight	Yellow	Block E	120	3	156
4	S2345	Guy	Richie	Green	Block D	45	2	26
5	S3434	Stephen	Williams	Green	Block D	45	2	34
#####						0	0	0

Staff and Garage are distinct entities

Staff and Garage are related in that a Staff member has a parking space in a garage

Garage, location, capacity, and floor have repeated values

3NF Form

Note no opportunity for update anomalies

- Staff table

ID	staffNo	staffFirstNam	staffLastNam	garageID	spaceNo
1	S1156	Jane	Jones	1	123
2	S2311	Karen	Gilmore	1	145
3	S1167	Richard	Blight	1	156
4	S2345	Guy	Richie	2	26
5	S3434	Stephen	Williams	2	34
(New)				0	0

- Garage table

ID	garageName	location	capacity	floor
1	Yellow	Block E	120	3
2	Green	Block D	45	2

Track Example

Assume surgeryNo means the number of the dental room

- Note the field multiple values
- Note that dentistName depends on staffNo
- Note that patName depends on patNo
- Note that surgeryNo depends on neither of the above

staffNo	dentistName	patNo	patName	Appointment		surgeryNo
S1011	Tony Smith	P100	Gillian White	12-Sep-08	10.00	S15
S1011	Tony Smith	P105	Jill Bell	12-Sep-08	12.00	S15
S1024	Helen Pearson	P108	Ian MacKay	12-Sep-08	10.00	S10
S1024	Helen Pearson	P108	Ian MacKay	14-Sep-08	14.00	S10
S1032	Robin Plevin	P105	Jill Bell	14-Sep-08	16.30	S15
S1032	Robin Plevin	P110	John Walker	15-Sep-08	18.00	S13

Track RDM

Dentists
firstName
lastName
Appointments
apDate
apTime
surgeryNo
Patients
firstName
lastName

Are We on Track?

An employee may have many positions in different organizations

- Using Access, convert the DB below to a normalized (3NF) Access DB
- Write a query to display the positions in each organization

orgName	posName
Pat & Co	IT Assistant
Will & Son	Asst. Researcher
Will & Son	Project Manager
Will & Son	Technician

eNum	eName	DOB	org	position	start	finish
0056	Kate Brown	12/12/76	Pat&Co	IT Assistant	11/1/90	4/30/95
0057	Gil Bradford	3/1/71	Will&Son	Technician	6/1/88	3/31/91
0057	Gil Bradford	3/1/71	Will&Son	Asst. Researcher	4/1/91	3/31/94
0057	Gil Bradford	3/1/71	Will&Son	Project Manager	4/1/94	1/31/96
0089	Steve Widden	9/9/83	Pat&Co	IT Assistant	3/1/91	8/31/95